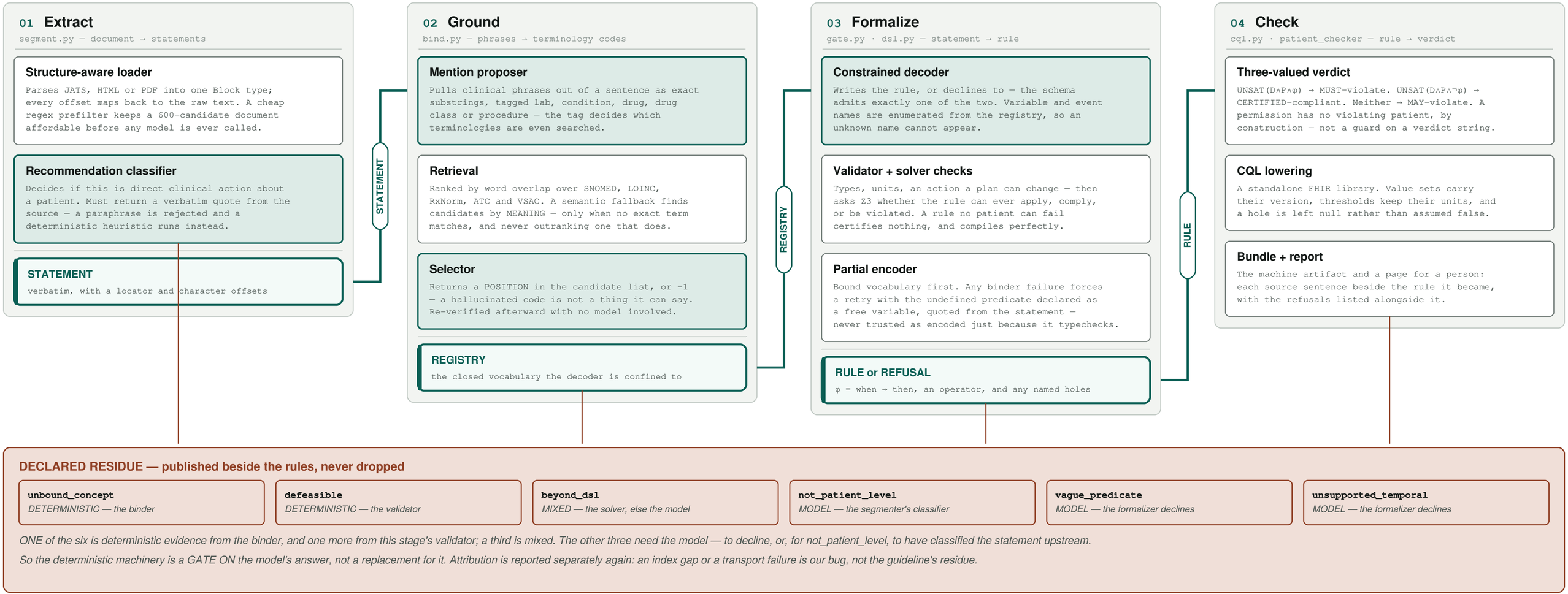

01 Extract
segment.py — document → statements
Structure-aware loader
Parses JATS, HTML or PDF into one Block type; every offset maps back to the raw text. A cheap regex prefilter keeps a 600-candidate document affordable before any model is ever called.
Recommendation classifier
Decides if this is direct clinical action about a patient. Must return a verbatim quote from the source — a paraphrase is rejected and a deterministic heuristic runs instead.
STATEMENT
verbatim, with a locator and character offsets
STATEMENT
02 Ground
bind.py — phrases → terminology codes
Mention proposer
Pulls clinical phrases out of a sentence as exact substrings, tagged lab, condition, drug, drug class or procedure — the tag decides which terminologies are even searched.
Retrieval
Ranked by word overlap over SNOMED, LOINC, RxNorm, ATC and VSAC. A semantic fallback finds candidates by MEANING — only when no exact term matches, and never outranking one that does.
Selector
Returns a POSITION in the candidate list, or -1 — a hallucinated code is not a thing it can say. Re-verified afterward with no model involved.
REGISTRY
the closed vocabulary the decoder is confined to
REGISTRY
03 Formalize
gate.py · dsl.py — statement → rule
Constrained decoder
Writes the rule, or declines to — the schema admits exactly one of the two. Variable and event names are enumerated from the registry, so an unknown name cannot appear.
Validator + solver checks
Types, units, an action a plan can change — then asks Z3 whether the rule can ever apply, comply, or be violated. A rule no patient can fail certifies nothing, and compiles perfectly.
Partial encoder
Bound vocabulary first. Any binder failure forces a retry with the undefined predicate declared as a free variable, quoted from the statement — never trusted as encoded just because it typechecks.
RULE or REFUSAL
φ = when → then, an operator, and any named holes
RULE
04 Check
cql.py · patient_checker — rule → verdict
Three-valued verdict
UNSAT(D∧P∧φ) → MUST-violate. UNSAT(D∧P∧¬φ) → CERTIFIED-compliant. Neither → MAY-violate. A permission has no violating patient, by construction — not a guard on a verdict string.
CQL lowering
A standalone FHIR library. Value sets carry their version, thresholds keep their units, and a hole is left null rather than assumed false.
Bundle + report
The machine artifact and a page for a person: each source sentence beside the rule it became, with the refusals listed alongside it.
DECLARED RESIDUE — published beside the rules, never dropped
unbound_concept
DETERMINISTIC — the binder
defeasible
DETERMINISTIC — the validator
beyond_dsl
MIXED — the solver, else the model
not_patient_level
MODEL — the segmenter's classifier
vague_predicate
MODEL — the formalizer declines
unsupported_temporal
MODEL — the formalizer declines
ONE of the six is deterministic evidence from the binder, and one more from this stage's validator; a third is mixed. The other three need the model — to decline, or, for not_patient_level, to have classified the statement upstream.
So the deterministic machinery is a GATE ON the model's answer, not a replacement for it. Attribution is reported separately again: an index gap or a transport failure is our bug, not the guideline's residue.
deterministic step
model call (4 total)
stage output → next input
declared residue
feeds the next stage
reaches the residue channel